

Project Design Phase-II - Solution Requirements

Date	Team ID	Project Name	Maximum Marks
18 June 2026		ShopEZ E-Commerce Platform	4 Marks

1. Functional Requirements (FR)

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Authentication	User registration, user login validation, session validation via JWT, local storage cleanup on token invalidation.
FR-2	Product Browsing	Keyword search bar queries, category-based rendering (Groceries, Accessories, Appliances), price-slider updates.
FR-3	Cart Management	Increment/decrement item quantity, calculate cart subtotals dynamically, verify inventory limits on updates.
FR-4	Order Checkout	Input shipping address, choose payment mode (COD, Card, UPI), display invoice details (Subtotal, GST, Delivery fees).
FR-5	Admin Access	Protected route authentication, product inventory catalog management (add, edit, delete).

2. Non-Functional Requirements (NFR)

NFR No.	Category	Requirement Description
NFR-1	Usability	The client must be responsive on mobile and desktop web browsers, with clean animations and a high-contrast dark mode design.
NFR-2	Security	Store user passwords securely using bcrypt hashing. JWT tokens must expire after 24 hours, and local storage must clear on auth errors.
NFR-3	Reliability	The database connection must auto-reconnect upon loss, and Express must handle errors gracefully without crashing (global errorHandler middleware).
NFR-4	Performance	Frontend search updates should execute in under 300ms, and API response times for product retrievals must be below 100ms.
NFR-5	Scalability	Support at least 1,000 concurrent database queries using MongoDB index keys on product names and categories.